

Summary: Performance of bent crystals for the TWOCRIST experiment

Introduction

The TWOCRIST experiment at the Large Hadron Collider (LHC) aims to demonstrate "double-channelling" of protons at TeV energies. This involves steering a particle beam through two consecutive bent crystals. If successful, this technique could enable the ALADDIN experiment, which proposes to measure the electric and magnetic dipole moments of short-lived charmed baryons (like Λ_c^+) by observing their spin precession within the intense electromagnetic fields of a long, bent crystal.

The experiment tests three silicon crystals to ensure they meet the stringent mechanical and performance requirements for the LHC.

Crystal Designs

The three crystals tested utilize different bending technologies:

1. **TCCS (Target Collimator Crystal for Splitting):** A short (4 mm) crystal with a small bend angle (50 μrad). It is bent using a U-shaped metallic holder, a well-established technique used for LHC beam collimation. Its purpose is to split protons from the main LHC beam halo.
2. **TCCP (Target Collimator Crystal for Precession):** A significantly longer (70 mm) crystal with a much larger bend angle (6.9 mrad). Like the TCCS, it uses a mechanically clamped metallic holder. Its purpose is to channel and induce spin precession in the charmed baryons.
3. **TCCPA (Target Collimator Crystal for Precession - Anodic):** A novel crystal of similar length (70.5 mm) but bent to an even larger angle (13.3 mrad). Instead of mechanical clamping, it is bent using an **anodic bonding** technique, where the silicon wafer is permanently bonded to a curved glass substrate using high voltage and temperature. This method was hypothesized to provide more uniform bending.

Characterisation with X-rays

Before beam testing, the crystals were examined using high-resolution X-ray diffraction (HR-XRD) to measure their bending radius and torsion.

- **TCCPA (Anodic Bonded):** Exhibited remarkable uniformity in its bending radius along its entire length.
- **TCCP (Mechanically Clamped):** Showed significant variation in its bending radius, particularly at the ends where the clamping forces are applied.

Characterisation with 180 GeV Hadron Beams

The crystals were tested at the CERN SPS H8 beamline using a 180 GeV secondary hadron beam (roughly 30% pions, 70% protons). The setup used high-precision goniometers to align the crystals and silicon microstrip detectors to track particle trajectories and deflection angles.

Analysis and Torsion Correction

A critical part of the analysis for the long crystals (TCCP and TCCPA) was correcting for **torsion** (τ_y)—the slight twisting of the crystallographic planes across the crystal's face. By mapping the optimal channeling angle across the crystal surface, the researchers found:

- **TCCP:** Torsion was relatively high ($24.4 \pm 0.2 \mu\text{rad/mm}$).
- **TCCPA:** Torsion was exceptionally low ($4.6 \pm 0.2 \mu\text{rad/mm}$), confirming the superior uniformity provided by the anodic bonding technique.

Measured Channelling Efficiency (ϵ_{ch})

Efficiency was calculated as the percentage of particles successfully channeled out of the total number of particles that entered the crystal within the acceptable angular range (the Lindhard angle).

- **TCCS:** $\epsilon_{ch} = 61.9 \pm 0.5\%$. This performance is excellent and aligns with existing collimation crystals.
- **TCCP:** $\epsilon_{ch} = 15.8 \pm 0.1\%$.
- **TCCPA:** $\epsilon_{ch} = 15.7 \pm 0.2\%$. Remarkably, the TCCPA achieved the same efficiency as the TCCP despite bending the beam nearly twice as much.

Simulations and Discrepancies

The experimental results were compared against Monte Carlo simulations using the Xsuite tracking code. The simulations assumed "perfect" crystals with uniform bending and no surface defects.

- **TCCS (Simulated: 74.8% vs. Measured: 61.9%):** The simulation slightly overestimated efficiency, likely due to idealized assumptions about detector resolution and minor crystal imperfections. However, the measured performance was deemed fully suitable for the LHC.
- **TCCP/TCCPA (Simulated: ~30-36% vs. Measured: ~16%):** There was a significant discrepancy between the simulated and measured efficiencies for the long crystals.

Conclusions on Discrepancies: The researchers attribute the lower-than-expected measured efficiency of the long crystals to physical inhomogeneities in the bending radius (which the simulation software currently cannot model). The fact that the TCCPA performed as well as the TCCP—despite a much sharper bend—supports the hypothesis that the highly uniform bending radius achieved via anodic bonding is crucial for maintaining efficiency in long crystals.

Final Conclusion

Both the TCCS and TCCP crystals were deemed suitable for installation in the LHC for the TWOCRIST experiment. While the long TCCP crystal had a lower absolute efficiency than idealized simulations predicted, it still demonstrated sufficient capability to perform the planned spin-precession experiments.

The novel anodic bonding technique used for the TCCPA proved highly successful at producing uniform bends with low torsion, highlighting its strong potential for future high-energy accelerator applications, though further testing regarding radiation tolerance is needed.

